Hüseyin Felek

R&D Engineer | Industrial Automation, Networking & Troubleshooting

Istanbul, Turkey | (+90) 505 974 84 01 | hseyin.felek@gmail.com | linkedin.com/in/hfelek

SUMMARY

Engineer with nearly 5 years in industrial automation and networking. Served as the technical escalation point for customer and field issues, running root cause analysis across software, communication and hardware, providing remote phone support, handling software licensing for products going to the field, and preparing technical documentation and training. Background in developing smart sensors, PLC-based automation controllers deployed on 1000+ field devices, and L2/L3 network features, with hands-on industrial communication (PROFINET, Modbus, IO-Link, RS-485) and TCP/IP networking.

WORK HISTORY

Software Design Engineer · Andasis Elektronik 05.2025 – 02.2026

- Stabilized core L2/L3 features (DHCP relay, IEEE 802.1X/RADIUS, ARP inspection, IGMP) on a Linux-based switching and routing platform, debugging each feature to production readiness.
- Configured VLAN, port mirroring and ACL rules, and root-caused reliability and throughput issues through packet-level analysis with Wireshark.
- Extended SNMP/MIB support for network monitoring.

Embedded Systems R&D Engineer · Eliar Elektronik 07.2021 – 05.2025

Field Support and Troubleshooting

- Served as the technical escalation point for customer and field issues on sensors and controllers, diagnosing problems across software, communication and hardware and responding to customers directly.
- Provided remote phone support to customers when they ran into problems with their products.
- Investigated serial and fieldbus communication issues (RS-485, Modbus, PROFINET) with logic analyzers, oscilloscopes and protocol analyzers, working with an international protocol stack vendor where needed.
- Handled CODESYS software licensing for products going to the field, completing and managing the licenses before delivery.
- Took part in first field trials, carried out post-delivery checks and resolved defects; led post-production quality assurance for the controller platform.

Training, Documentation and Customer Projects

- Trained the field service team on calibration procedures and supported them on technical questions.
- Wrote production test procedures, manufacturing guidelines and technical documentation, and handed tests and calibration procedures over to the production team.
- Led a 6-month on-site feasibility study at customer plants: interviewed process teams, analyzed data with Python, SQL and Power BI, and presented the findings to management.

Industrial Communication and Software

- Developed I/O module drivers and data exchange over PROFINET and Modbus for an embedded Linux automation controller platform, from a PROFINET I/O coupler to a full CODESYS PLC.
- Integrated the CODESYS runtime into controller products and developed IEC 61131-3 applications.
- Led the shared Bluetooth and Wi-Fi communication stack for the sensor line and integrated TCP/IP, UDP, CAN, RS-485 and IO-Link interfaces.
- Built web-based interfaces for device configuration, monitoring and OTA updates, along with REST and MQTT data interfaces for web and mobile applications.
- Managed device images and releases with versioned OTA updates, backup and restore, and Docker-based deployment of on-device services.
- Redesigned the controllers' I/O architecture from full-cycle to change-based updates and rolled it out to 1000+ field devices, replacing legacy controllers.

Software-Hardware Integration

- Developed real-time STM32/FreeRTOS firmware for smart sensors with a customized secure bootloader (SBSFU) for over-the-air updates, and implemented power-backup logic to protect controllers and data against unexpected power loss.

- Defined requirements for a new dual-channel pH/ORP product and designed its analog front-end for high-impedance measurement.
- Built CODESYS-based and PC-based test and calibration systems with automated test sequencing and data logging.
- Led certification testing for the company's first BLE-certified device.

CORE COMPETENCIES

Troubleshooting & Support	Root cause analysis, field and remote phone support, software licensing, customer communication, technical training, documentation (Confluence), Jira
Industrial Communication	PROFINET IO, Modbus RTU/TCP, IO-Link, RS-485, RS-232, CAN, MQTT, Bluetooth, Wi-Fi
Networking	TCP/IP, UDP, L2/L3 switching, VLAN, IEEE 802.1X/RADIUS, ACL, SNMP, Wireshark
Software & Platforms	CODESYS (IEC 61131-3), C/C++, Python, Bash, embedded Linux, Docker, REST APIs
Hardware & Instrumentation	STM32, ESP32, FreeRTOS, analog front-end design, 4-20 mA, oscilloscopes, logic and protocol analyzers
Data & Reporting	Excel, Power BI, SQL, Grafana

EDUCATION & LANGUAGES

Bachelor of Science, Electrical and Electronics Engineering · Boğaziçi University 2016 – 2021

Turkish Native | English Advanced | French Intermediate